RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF Sync-Inventory Guardrail Remediation & Host-Network Conformance

Clearing the domain guardrail pre-checks that abort a brownfield Sync Inventory — only ERRORs block; the DRS VmotionRate inversion; safely, protecting a healthy vSAN.

v1.1 VMware Cloud Foundation 9

INTERNAL

VCF Sync-Inventory Guardrail Remediation & Host-Network Conformance

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Platform:** VMware Cloud Foundation 9.0.1 (brownfield mgmt domain `95727d6d…`, cluster `vcenter-cl01`)

> **When to use.** A **Sync Inventory** (brownfield reconciliation) aborts because domain **guardrail pre-checks fail** — `DomainSyncException: … has failed guard rails`. This runbook clears the failing guardrails **safely** (protecting a healthy vSAN) so the sync can proceed. Companion to [[RUNBOOK-SDDC-Host-Inventory-Desync-and-Sync-Inventory-Brownfield-2026-07-27]].

1. How to Read the Guardrail Result

The brownfield tool writes per-run output on **SDDC Manager** (readable as `vcf`, no sudo):

```
/var/log/vmware/vcf/domainmanager/brownfield/<run-id>/vcf_brownfield.log        # narrative + the abort traceback
/var/log/vmware/vcf/domainmanager/brownfield/<run-id>/output/guardrails_report_<vc>_all.csv   # every check + Status + Remediation
/var/log/vmware/vcf/domainmanager/brownfield/<run-id>/output/results_mgmt.json    # STARTED / FAIL summary
```

Use the run whose CSV has **data rows** (a header-only 124-byte CSV = a clobbered duplicate). Grep the `_all.csv` for `VALIDATION_FAILED`. The columns are: *Object Type, Object Name, Day-N Operation, Severity, Status, Check Name, Description, Details, Remediation.*

> **`set datasource: 400` is a RED HERRING.** The log shows `DATA_SOURCE_ALREADY_CONFIGURED` right after it — the tool logs the 400 and continues. **The real blocker is always the failed guardrails**, not the datasource line.

2. Verified State (2026-07-27) — the baseline before any change

vSAN is HEALTHY — one 4-node cluster; every host reports the same Sub-Cluster UUID, Member Count 4, all four hostnames. **No partition, no data risk.** Confirm before *and* after any change:

```
esxcli vsan cluster get | grep -E "Sub-Cluster Member Count|Local Node State"   # want Count 4
esxcli vsan health cluster get -w 2>/dev/null | head        # or: check vSAN health in vCenter
```

Network map (by service TAG — not vmk number):

Host	vSAN	vMotion	Note
esxi01	vmk1 192.168.12.121	vmk2 192.168.11.121	**vmk1 also tagged `Management` ← the defect**
esxi02	vmk1 192.168.12.124	vmk2 192.168.11.124	conformant
esxi03	**vmk2** 192.168.12.122	**vmk1** 192.168.11.122	vmk numbers reversed (cosmetic)
esxi04	vmk1 192.168.12.123	vmk2 192.168.11.123	conformant

By subnet all four agree: **vSAN = 192.168.12.x, vMotion = 192.168.11.x.** No duplicate IPs.

3. The Guardrail Failures & Fixes

Confirmed 2026-07-27: only `ERROR` -severity guardrails block the sync — `WARNING` s do NOT. A `Sync domain mgmt` run completed `status: PASS` with the upgrade-policy **WARNING still unresolved.** So **fix the ERRORs; the WARNING is optional.** In practice the ONLY change actually required here was the esxi01 tag (§3.1) — DRS was already satisfied (§3.3).

3.1 esxi01 — more than one Management/vMotion vmknic (ERROR) — *the one required fix*

esxi01 has `Management` on **both** vmk0 and vmk1 (`vmk1 -> VSAN, Management`). Remove the stray tag from vmk1 (management stays on vmk0; vSAN stays on vmk1):

```
 # On esxi01 (verify first, then remove, then verify)
esxcli network ip interface tag get -i vmk1                    # shows: VSAN, Management
esxcli network ip interface tag remove -i vmk1 -t Management   # remove ONLY the Management tag
esxcli network ip interface tag get -i vmk1                    # should now show: VSAN
esxcli vsan network list                                       # vmk1 still vSAN — unchanged
```

Risk: low — you are removing an *extra* tag, not an interface or IP. **Rollback:** `esxcli network ip interface tag add -i vmk1 -t Management`. Confirm esxi01 stays reachable on vmk0 and vSAN Member Count stays 4.

3.2 vcenter-cl01 — ESXi upgrade policy mismatch (WARNING) — **OPTIONAL, does NOT block**

The cluster's ESXi upgrade policy (**"Evacuate Offline VMs"**) does not match the SDDC default. **This is a WARNING — the sync PASSES with it unresolved (confirmed 2026-07-27), so fixing it is optional cleanliness, not required.** If you choose to align it: vCenter UI → Cluster `vcenter-cl01` → **Configure → vSphere Lifecycle Manager / cluster upgrade settings** → match the SDDC default (verify the exact value on-screen; don't guess). Risk: low. Rollback: restore the prior policy.

3.3 vcenter-cl01 — DRS automation (ERROR) — *usually already satisfied*

The guardrail checks that DRS **Automation Level = Fully Automated**. On this cluster it **already was**, so this ERROR was effectively already cleared — **verify before changing anything.**

CRITICAL — the DRS `VmotionRate` (migration threshold) is INVERSE of the UI slider. In the API/PowerCLI, `VmotionRate=5` **= UI *Conservative*** (fewest moves) and `VmotionRate=1` **= UI *Aggressive*** (most moves). Writing `VmotionRate=1` intending "conservative" actually makes DRS **maximally aggressive** and triggers vMotions (learned the hard way 2026-07-27). The **migration threshold is separate from this guardrail** — the guardrail only cares about the *automation level*, not the threshold. So **leave the threshold at your preferred setting** (Conservative = `VmotionRate=5` for a fragile nested lab).

- **Check first (read-only):** `Get-Cluster <name> | Select DrsAutomationLevel; (Get-Cluster <name>).ExtensionData.ConfigurationEx.DrsConfig.VmotionRate`.
- **If** the level is genuinely not Fully Automated: vCenter UI → Cluster → **Configure** → **vSphere DRS** → **Edit** → **Automation Level = Fully Automated** → OK. **Set the threshold via the UI slider** (visually unambiguous) rather than the inverted API value.

4. What NOT to Change

Do not re-IP or re-number any vSAN/vMotion vmknic to "fix" esxi03. esxi03's vmk numbering is reversed (vmk1=vMotion, vmk2=vSAN) but **by tag it is on the correct subnets and vSAN is healthy.** Re-IPing a live vSAN interface can partition the host and take objects offline. Only undertake vmknic re-numbering/re-IP if a **documented network design** requires it — and then as a **separate, per-host, maintenance-mode-evacuated** project with vSAN health verified at each step, HA disabled for the window, and a full backup. It is **not** part of this guardrail fix.

5. Execution Order & Safety Gates

1. **Baseline:** vSAN health = 4 members, no resync backlog (`esxcli vsan debug object health summary get`), note current DRS level.
2. **Fix 3.1** (esxi01 tag) → re-verify vSAN Count 4 + esxi01 reachable.
3. **Fix 3.2** (upgrade policy).
4. **Fix 3.3** (DRS Fully Automated) → watch vMotion activity briefly.
5. **Drain any hung sync tasks → run [[RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup-2026-06-09]]** (gate: 0 `IN_PROGRESS`) to clear the lock/task jam.
6. **Re-run ONE Sync Inventory** (single click, ignore the 504 — never retry in the UI).
7. **Verify:** the new brownfield run's `results_mgmt.json` → `status: PASS` (or guardrails all green), then `/v1/hosts` shows esxi02/03/04 flip to `ASSIGNED`.
8. **(Optional)** revert DRS to Partially Automated if that is your steady state.

6. If Guardrails Still Fail

Read the fresh run's `guardrails_report_*_all.csv` for any remaining `VALIDATION_FAILED` rows and apply each row's **Remediation** column. The report is authoritative and self-service — **no Broadcom support required.**

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1.0	2026-07-27	Initial runbook.
1.1	2026-07-27	Corrected from the confirmed 2026-07-27 resolution: only ERROR guardrails block (a run PASSED with the upgrade-policy WARNING unresolved); DRS was already Fully Automated; documented the DRS VmotionRate inversion (5=Conservative, 1=Aggressive) after a mis-set briefly triggered vMotions.

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